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HEARING, SMELL AND TASTE

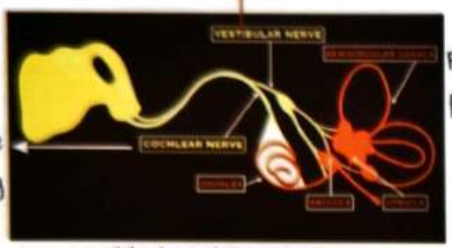
----- Active space --

Hearing

00:00:17

COCHLEA :

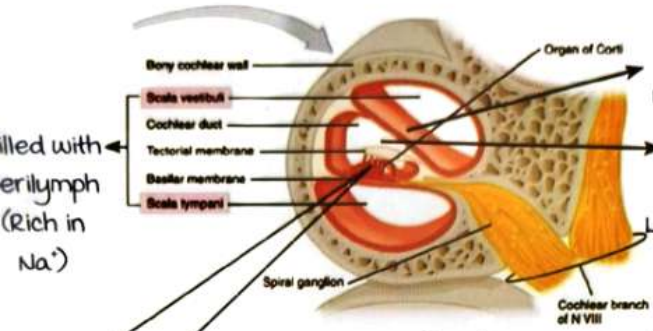
Responsible for maintaining balance



Responsible for hearing

Vestibulocochlear nerve (8th cranial nerve)

Filled with perilymph (Rich in Na^+)



Parts of cochlea

Reissner's memerane

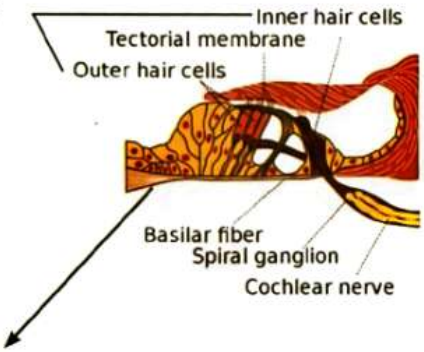
Scala media

Lined with stria vascularis

Secrets K^+ rich endolymph (Endolymphatic potential : + 80 mv)

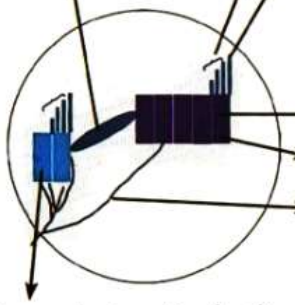
Organ of Corti :

Senses sound & relays information to spinal ganglion



Hair Cells :

Deiter hair cell (Supporting cell)



Cilia

Tallest : Kinocilium

Shorter ones: Stereocilia

Outer hair cells (OHC) :

- more in number.
- Less innervated.

Inner hair cells (IHC) :

- Less in number.
- more innervated.

Basilar membrane :

AKA frequency analyzer.

Principle : Travelling wave theory.

Base	Apex
Broad	Narrow
Close to oval window	Close to helicotrema
↑ Vibration by high frequency waves	↑ Vibration by low frequency waves

Hair cell movement (In response to sound)

- Towards Kinocilium → K^+ influx (From endolymph) → Depolarization → Conducts nerve impulse
- Away from Kinocilium → Hyperpolarization → Inhibits nerve conduction

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Applied aspect :

1. Hair cells : Prone to damage (Ototoxicity).
   - Outer > inner.
   - Commonly by drugs (Aminoglycosides).
2. Otoacoustic emissions (OAE).
   - K^+ current measured in OHCs (Kemp waves).
   - Sensitive hearing test.

NEURAL PATHWAY :

Auditory pathway :

Mnemonic: E COLI MA.

Eighth nerve
↓
Cochlear nuclei
↓
Superior olivary nucleus
↓
Lateral lemniscus
↓
Inferior colliculus
↓
Medial geniculate body
↓
Auditory cortex

Detects direction of sound source

Auditory cortex :
Higher order processing center for hearing.



Brodmann area : 41, 42

- Superior temporal gyrus/heschl's gyrus/ planum temporale.

Stapedial reflex :

AKA attenuation reflex/preventive reflex

Loud noise
↓
8th cranial nerve (Afferent)
↓
Superior olivary nucleus (Center)
↓
Facial nerve (Efferent)
↓
Contraction of stapedius
↓
Attenuates sound
↓
Prevents rupture of tympanic membrane

Note :

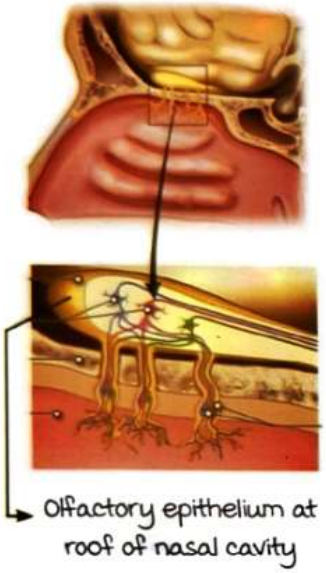
Facial nerve paralysis → Stapedius does not contract → Hyperacusis (Loud noise perceived louder).

Feedback

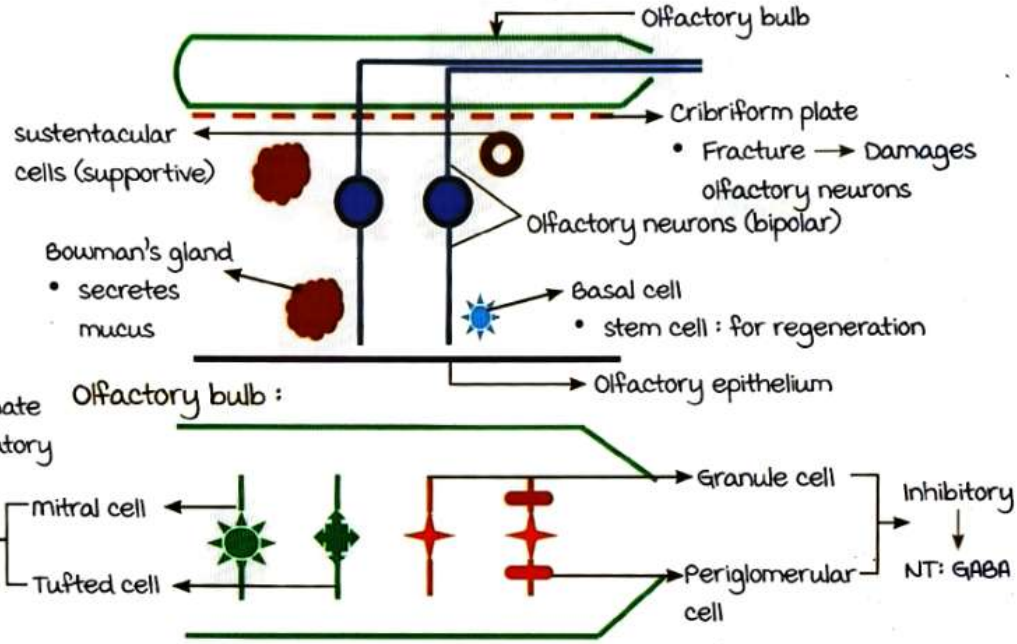
Smell

00:29:40 ----- Active space -----

OLFACTORY PATHWAY :



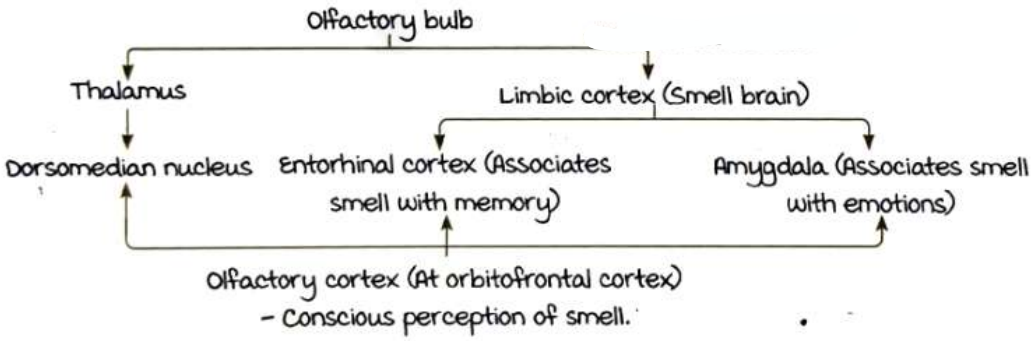
Olfactory epithelium at roof of nasal cavity



NT: Glutamate ↑ Excitatory

Olfactory bulb :

Olfaction :



Note :

Entorhinal cortex : Earliest area to undergo degeneration in Alzheimer's.

Taste (Gustation)

00:42:00

Taste buds :

Taste	Receptors
Sweet	TIR (G protein coupled receptors)
Salt	Epithelial Na channels (ENaC)
Sour	Hyperpolarisation cyclic nucleotide (HCN) channel
Bitter – helps in survival (most toxins : Bitter)	T2 receptors
Umami – Highly addictive (In ripe tomatoes)	m glutamate receptors 4 (mGLUTR4)

Feedback

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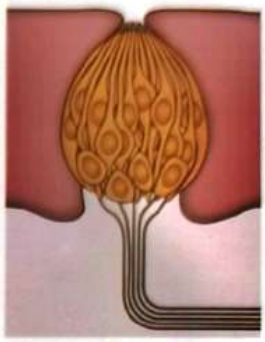
Cells :

Type I : Glial cells.

Type II : Taste receptor cells.

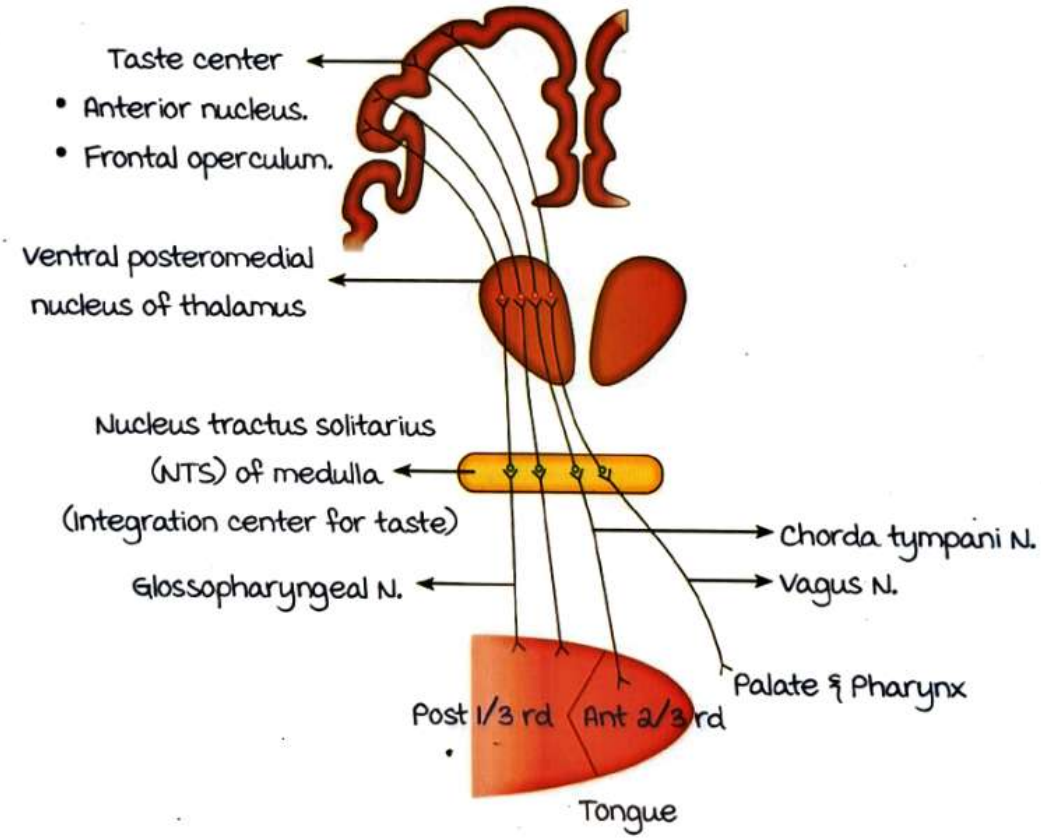
Type III : Presynaptic cell.

Type IV : Basal cell (Stem cell).



Taste bud : Cells

TASTE PATHWAY :



Feedback